

The University of Texas at Austin
IN-CLASS WORK 8

Introduction to Financial Mathematics

September 25, 2026

HEDGING USING EUROPEAN PUT OPTIONS.

Problem 8.1. Farmer Shaun is producing sweet potatoes. He intends to harvest 10,000 —cartons' worth in six months. His total costs are \$12.00 per carton.

He wishes to hedge using European put options. There are two puts on sweet potatoes with the exercise date in six months available: one with the strike price of \$13 per carton and another with the strike price of \$15 per carton. Their premiums are \$0.15 and \$0.18, respectively.

Assume that the prevailing risk-free interest rate is 4% effective for the half-year period.

At harvest time, in six months, it turns out that the sweet-potato spot price equals \$14. What would Farmer Shaun's profit be if he had decided to hedge using the \$13-strike put versus his profit if he had decided to use the \$15-strike put to hedge?

Solution. Farmer Shaun's unhedged position has the following profit:

$$10,000(S(T) - 12)$$

where $S(T)$ stands for the spot price of sweet potatoes in six months.

If he decided to hedge using put options, he would **long** the put. So, the profit of the \$13-strike-put hedge would be:

$$10,000(13 - S(T))_+ - 10,000 \times 0.15 \times 1.04.$$

The profit of the \$15-strike-put hedge would be:

$$10,000(15 - S(T))_+ - 10,000 \times 0.18 \times 1.04.$$

The profit of the hedged position with the given $S(T) = 14$ in the first case equals

$$10,000(14 - 12 - 0.15 \times 1.04) = 18,440.$$

For the second insurance strategy, the profit is

$$10,000(14 - 12 + (15 - 14) - 0.18 \times 1.04) = 28,128.$$

Problem 8.2. Aunt Dahlia simultaneously purchased

- one share of a market index at the current spot price of \$1,000;
 - one one-year, \$1,050-strike put option on the above market index for the premium of \$20.
- i. Is the above portfolio's payoff bounded from above? If you believe it is not, substantiate your claim. If you believe that it is, provide the upper bound.
 - ii. Is the above portfolio's payoff bounded from below? If you believe it is not, substantiate your claim. If you believe that it is, provide the lower bound.

Solution. The payoff of the portfolio expressed in terms of the final asset price $S(T)$ is

$$V(T) = S(T) + (K - S(T))_+ = \max[K, S(T)] = K \vee S(T).$$

- i. The payoff is **not** bounded from above since the stock price $S(T)$ may be arbitrarily large.
- ii. The payoff is bounded from below by the put option's exercise price K . This means that there is a guarantee of the minimum price the owner of the portfolio can fetch for the underlying asset. That's why this type of a portfolio is referred to as the *floor*.

COVERED PUTS.

Problem 8.3. Trader Mavis shorts 10,000 shares of XYZ stock at the current price of \$50.00 per share, to be returned in six months. She does not invest the proceeds of the initial sale.

She wishes to enhance her bearish position using a covered put. There are two European put options on XYZ stock with the exercise date in six months available: one with the strike price of \$48 per share and another with the strike price of \$45 per share. Their premiums are \$1.20 and \$0.80, respectively.

Assume that the prevailing risk-free interest rate is 2% effective for the half-year period.

In six months, it turns out that XYZ's spot price equals \$46. What would Trader Mavis's profit be if she had decided to write the \$48-strike put versus her profit if she had decided to write the \$45-strike put, in each case to cover her short position?

Solution. Trader Mavis's unhedged short position has the following profit:

$$10,000(50 - S(T))$$

where $S(T)$ stands for the spot price of XYZ stock in six months.

If she decided to enhance the position with a covered put, she would additionally short the put. So, the profit of the \$48-strike covered put would be:

$$10,000(50 - S(T)) - 10,000(48 - S(T))_+ + 10,000 \times 1.20 \times 1.02.$$

The profit of the \$45-strike covered put would be:

$$10,000(50 - S(T)) - 10,000(45 - S(T))_+ + 10,000 \times 0.80 \times 1.02.$$

With the given $S(T) = 46$, in the first case the profit equals

$$10,000(50 - 46) - 10,000(48 - 46) + 10,000 \times 1.20 \times 1.02 = 32,240.$$

For the second strategy, the profit is

$$10,000(50 - 46) - 0 + 10,000 \times 0.80 \times 1.02 = 48,160.$$

Problem 8.4. Uncle Baxter simultaneously

- sold short one share of a market index at the current spot price of \$1,000, to be returned in one year;
 - wrote one one-year, \$900-strike put option on the same market index, receiving a premium of \$30.
- i. Is the above portfolio's payoff bounded from above? If you believe it is not, substantiate your claim. If you believe that it is, provide the upper bound.

- ii. Is the above portfolio's payoff bounded from below? If you believe it is not, substantiate your claim. If you believe that it is, provide the lower bound.

Solution. The payoff of the portfolio expressed in terms of the final asset price $S(T)$ is

$$V(T) = -S(T) - (K - S(T))_+ = -\max[K, S(T)].$$

- i. The payoff is bounded from above, by $-K = -900$. Whenever $S(T) \leq K$, the payoff equals exactly $-K$, and it cannot exceed this value.
- ii. The payoff is **not** bounded from below, since the index price $S(T)$ may be arbitrarily large, and the short-stock leg then produces an arbitrarily large loss. Unlike the covered call, which caps losses on a long position, a covered put only caps the *profit* of a short position — the risk of the short-stock leg remains unlimited to the upside.

HEDGING USING A LONG PUT AND A SHORT CALL.

Problem 8.5. Grower Percival is producing almonds. He intends to harvest 8,000 pounds' worth in six months. His total costs are \$2.50 per pound.

He wishes to hedge his position using one long options position and one short options position on almonds, both with exercise date in six months. He considers two strategies:

1. Buy a put with strike price \$3.00 per pound (premium \$0.20) and sell a call with strike price \$3.50 per pound (premium \$0.10).
2. Buy a put with strike price \$2.80 per pound (premium \$0.10) and sell a call with strike price \$3.20 per pound (premium \$0.12).

Assume that the prevailing risk-free interest rate is 2.5% effective for the half-year period.

At harvest time, in six months, it turns out that the almond spot price equals \$3.55. What would Grower Percival's profit be under the first strategy versus his profit under the second strategy?

Solution. Grower Percival's unhedged position has the following profit:

$$8,000(S(T) - 2.50)$$

where $S(T)$ stands for the spot price of almonds in six months.

Under the first strategy, he is long the \$3.00-strike put and short the \$3.50-strike call. So, the profit of the first strategy would be:

$$8,000(S(T) - 2.50) + 8,000(3.00 - S(T))_+ - 8,000(S(T) - 3.50)_+ - 8,000 \times 0.10 \times 1.025.$$

The profit of the second strategy, long the \$2.80-strike put and short the \$3.20-strike call, would be:

$$8,000(S(T) - 2.50) + 8,000(2.80 - S(T))_+ - 8,000(S(T) - 3.20)_+ - 8,000 \times (-0.02) \times 1.025.$$

Note that the second strategy involves a *net premium credit* of \$0.02 per pound, since the call premium received exceeds the put premium paid.

The profit of the hedged position under the first strategy, with the given $S(T) = 3.55$, equals

$$8,000(3.55 - 2.50) + 0 - 8,000(3.55 - 3.50) - 8,000 \times 0.10 \times 1.025 = 7,180.$$

For the second strategy, the profit is

$$8,000(3.55 - 2.50) + 0 - 8,000(3.55 - 3.20) + 8,000 \times 0.02 \times 1.025 = 5,764.$$

Problem 8.6. Cousin Reginald simultaneously purchased

- one share of a market index at the current spot price of \$1,000;
 - one one-year, \$900-strike put option on the above market index for the premium of \$15;
 - and also wrote one one-year, \$1,100-strike call option on the same market index, receiving a premium of \$20.
- i. Is the above portfolio's payoff bounded from above? If you believe it is not, substantiate your claim. If you believe that it is, provide the upper bound.
 - ii. Is the above portfolio's payoff bounded from below? If you believe it is not, substantiate your claim. If you believe that it is, provide the lower bound.

Solution. The payoff of the portfolio expressed in terms of the final asset price $S(T)$ is

$$V(T) = S(T) + (900 - S(T))_+ - (S(T) - 1,100)_+.$$

Consider the three ranges of $S(T)$ separately: if $S(T) \leq 900$, then $V(T) = 900$ if $900 < S(T) \leq 1,100$, then $V(T) = S(T)$ and if $S(T) > 1,100$, then $V(T) = 1,100$.

- i. The payoff **is** bounded from above, by \$1,100 — the strike of the written call. Once the index rises past \$1,100, the shares are called away at that price, so the payoff can never exceed it.
- ii. The payoff **is** bounded from below, by \$900 — the strike of the purchased put. Once the index falls below \$900, the put may be exercised, guaranteeing a sale at that price, so the payoff can never fall below it.

HEDGING USING A SHORT PUT AND A LONG CALL.

Problem 8.7. Confectioner Wilhelmina is producing chocolate and will need to purchase 6,000 pounds of cocoa in six months to fulfill her production needs.

She wishes to hedge against a rise in the price of cocoa using a long call together with a short put, both with exercise date in six months. She considers two strategies:

1. Buy a call with strike price \$4.00 per pound (premium \$0.25) and sell a put with strike price \$3.50 per pound (premium \$0.15).
2. Buy a call with strike price \$4.20 per pound (premium \$0.15) and sell a put with strike price \$3.70 per pound (premium \$0.20).

Assume that the prevailing risk-free interest rate is 3% effective for the half-year period.

In six months, it turns out that the cocoa spot price equals \$4.50. What would Confectioner Wilhelmina's effective cost of acquiring the cocoa be under the first strategy versus the second strategy?

Solution. Confectioner Wilhelmina's unhedged cost of acquiring the cocoa is

$$6,000 \times S(T)$$

where $S(T)$ stands for the spot price of cocoa in six months.

Under the first strategy, she is long the \$4.00-strike call and short the \$3.50-strike put. So, the cost of the first strategy would be:

$$6,000 \times S(T) - 6,000(S(T) - 4.00)_+ + 6,000(3.50 - S(T))_+ + 6,000 \times 0.10 \times 1.03.$$

The cost of the second strategy, long the \$4.20-strike call and short the \$3.70-strike put, would be:

$$6,000 \times S(T) - 6,000(S(T) - 4.20)_+ + 6,000(3.70 - S(T))_+ + 6,000 \times (-0.05) \times 1.03.$$

Note that the second strategy involves a *net premium credit* of \$0.05 per pound, since the put premium received exceeds the call premium paid.

The cost under the first strategy, with the given $S(T) = 4.50$, equals

$$6,000 \times 4.50 - 6,000(4.50 - 4.00) + 0 + 6,000 \times 0.10 \times 1.03 = 24,618.$$

For the second strategy, the cost is

$$6,000 \times 4.50 - 6,000(4.50 - 4.20) + 0 - 6,000 \times 0.05 \times 1.03 = 24,891.$$

So, even though the second strategy is financed with a net premium credit, its higher call strike leaves her exposed to more of the price increase, making it the more expensive hedge in this outcome.